



High Quality Audio , J-FET Input, Dual Operational Amplifier

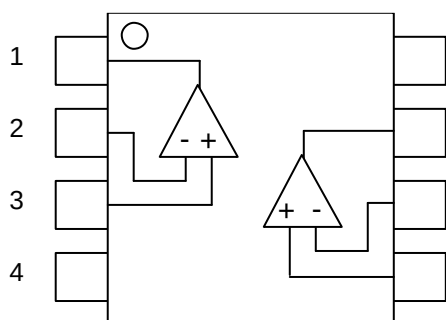
The **MUSES01** is a dual J-FET input high quality audio operational amplifier, which is optimized for high-end audio and professional audio applications with advanced circuitry and layout, unique material and assembled technology by skilled-craftwork.

It is the best for audio preamplifiers, active filters, and line amplifiers with excellent sound.

■ FEATURES

•Operating Voltage	$V_{opr} = \pm 9V$ to $\pm 16V$
•Output noise	$9.5nV/\sqrt{Hz}$ at $f=1kHz$
•Input Offset Voltage	$0.8mV$ typ. $5mV$ max.
•Input Bias Current	$200pA$ typ. $800pA$ max. at $T_a=25^{\circ}C$
•Voltage Gain	$105dB$ typ.
•Slew Rate	$12V/\mu s$ typ.
•Bipolar Technology	
•Package Outline	DIP8

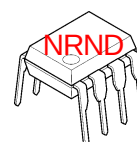
■ PIN CONFIGURATION



PIN FUNCTION

1.	A OUTPUT
2.	A -INPUT
3.	A +INPUT
4.	V-
5.	B +INPUT
6.	B -INPUT
7.	B OUTPUT
8.	V+

■ PACKAGE OUTLINE



MUSES01



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■ ABSOLUTE MAXIMUM RATINGS ($T_a=25^{\circ}C$)

PARAMETER	SYMBOL	RATING	UNIT
Supply Voltage	V^+/V^-	± 18	V
Common Mode Input Voltage	V_{ICM}	± 15 (Note1)	V
Differential Input Voltage	V_{ID}	± 30	V
Power Dissipation	P_D	910	mW
Output Current	I_O	± 25	mA
Operating Temperature Range	T_{opr}	-40 to +85	°C
Storage Temperature Range	T_{stg}	-50 to +150	°C

(Note1) For supply Voltages less than ± 15 V, the maximum input voltage is equal to the Supply Voltage.

■ RECOMMENDED OPERATING CONDITION ($T_a=25^\circ\text{C}$)

PARAMETER	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Supply Voltage	V^+/V^-	-	± 9	-	± 16	V

■ ELECTRIC CHARACTERISTICS

DC CHARACTERISTICS ($V^+/V^-=\pm 15\text{V}$, $T_a=25^\circ\text{C}$ unless otherwise specified)

PARAMETER	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Operating Current	I_{cc}	No Signal, $R_L=\infty$	-	8.5	12.0	mA
Input Offset Voltage	V_{IO}	$R_s \leq 10\text{k}\Omega$ (Note2)	-	0.8	5.0	mV
Input Bias Current	I_B	(Note2, 3)	-	200	800	pA
Input Offset Current	I_{IO}	(Note2, 3)	-	100	400	pA
Voltage Gain	A_v	$R_L \geq 2\text{k}\Omega$, $V_o = \pm 10\text{V}$	90	105	-	dB
Common Mode Rejection Ratio	CMR	$V_{ICM} = \pm 8\text{V}$ (Note4)	60	75	-	dB
Supply Voltage Rejection Ratio	SVR	$V^+/V^-=\pm 9.0$ to $\pm 16.0\text{V}$ (Note2, 5)	70	83	-	dB
Max Output Voltage 1	V_{OM1}	$R_L = 10\text{k}\Omega$	± 12	± 13.5	-	V
Max Output Voltage 2	V_{OM2}	$R_L = 2\text{k}\Omega$	± 10	± 12.5	-	V
Input Common Mode Voltage Range	V_{ICM}	CMR $\geq 60\text{dB}$	± 8	± 9.5	-	V

(Note2) Measured at $V_{ICM}=0\text{V}$

(Note3) Written by the absolute rate.

(Note4) CMR is calculated by specified change in offset voltage. ($V_{ICM}=0\text{V}$ to $+8\text{V}$ and $V_{ICM}=0\text{V}$ to -8V)

(Note5) SVR is calculated by specified change in offset voltage. ($V^+/V^-=\pm 9\text{V}$ to $\pm 16\text{V}$)

AC CHARACTERISTICS ($V^+V^- = \pm 15V$, $T_a = 25^\circ C$ unless otherwise specified)

PARAMETER	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Gain Bandwidth Product	GB	$f = 10kHz$	-	3.3	-	MHz
Unity Gain Frequency	f_T	$A_V = +100$, $R_S = 100\Omega$, $R_L = 2k\Omega$, $C_L = 10pF$	-	3.0	-	MHz
Phase Margin	ϕ_M	$A_V = +100$, $R_S = 100\Omega$, $R_L = 2k\Omega$, $C_L = 10pF$	-	60	-	deg
Input Noise Voltage1	V_{NI}	$f = 1kHz$, $A_V = +100$, $R_S = 100\Omega$	-	9.5	-	nV/ $\sqrt{Hz}$
Input Noise Voltage2	V_{N2}	RIAA, $R_S = 2.2k\Omega$, 30kHz LPF	-	1.2	3.0	μV_{rms}
Total Harmonic Distortion	THD	$f = 1kHz$, $A_V = +10$, $R_L = 2k\Omega$, $V_o = 5V_{rms}$	-	0.002	-	%
Channel Separation	CS	$f = 1kHz$, $A_V = +100$, $R_S = 1k\Omega$, $R_L = 2k\Omega$	-	150	-	dB
Positive Slew Rate	+SR	$A_V = 1$, $V_{IN} = 2V_{pp}$, $R_L = 2k\Omega$, $C_L = 10pF$	-	12	-	V/ μs
Negative Slew Rate	-SR	$A_V = 1$, $V_{IN} = 2V_{pp}$, $R_L = 2k\Omega$, $C_L = 10pF$	-	13	-	V/ μs

■ Application Notes

•Package Power, Power Dissipation and Output Power

IC is heated by own operation and possibly gets damage when the junction power exceeds the acceptable value called Power Dissipation P_D . The dependence of the MUSES01 P_D on ambient temperature is shown in Fig 1. The plots are depended on following two points. The first is P_D on ambient temperature 25°C, which is the maximum power dissipation. The second is 0W, which means that the IC cannot radiate any more. Conforming the maximum junction temperature T_{jmax} to the storage temperature T_{stg} derives this point. Fig.1 is drawn by connecting those points and conforming the P_D lower than 25°C to it on 25°C. The P_D is shown following formula as a function of the ambient temperature between those points.

$$\text{Dissipation Power } P_D = \frac{T_{jmax} - T_a}{\theta_{ja}} \text{ [W]} \quad (T_a = 25^\circ\text{C to } T_a = 150^\circ\text{C})$$

Where, θ_{ja} is heat thermal resistance which depends on parameters such as package material, frame material and so on. Therefore, P_D is different in each package.

While, the actual measurement of dissipation power on MUSES01 is obtained using following equation.

$$(\text{Actual Dissipation Power}) = (\text{Supply Voltage } V_{DD}) \times (\text{Supply Current } I_{DD}) - (\text{Output Power } P_o)$$

The MUSES01 should be operated in lower than P_D of the actual dissipation power.

To sustain the steady state operation, take account of the Dissipation Power and thermal design.

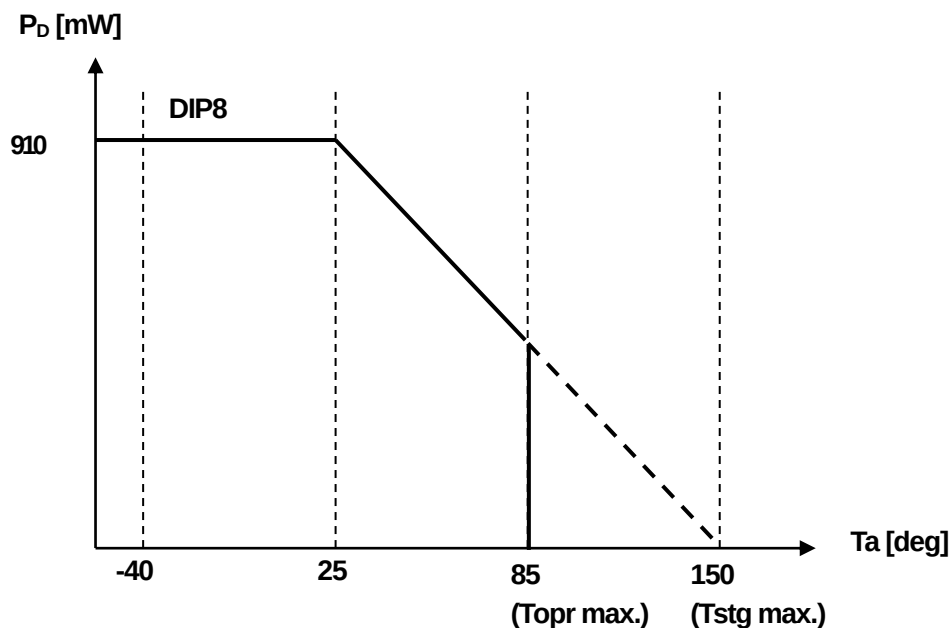
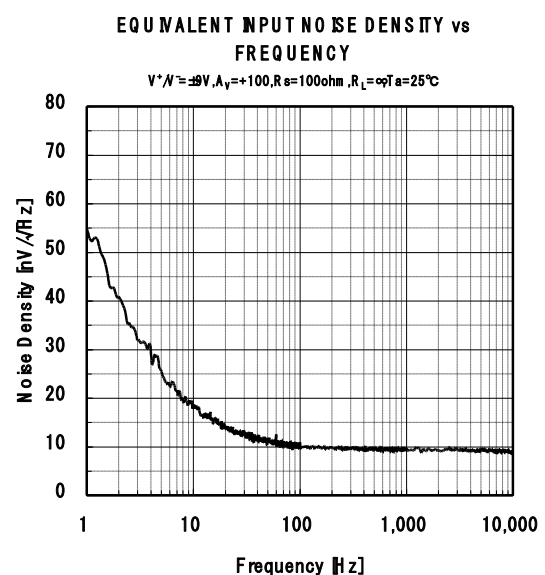
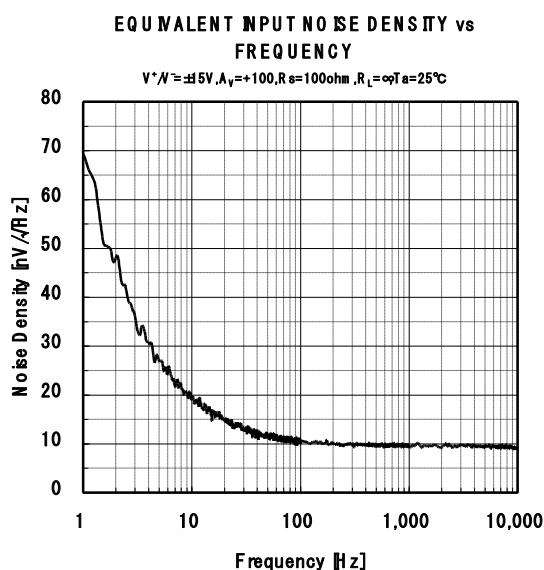
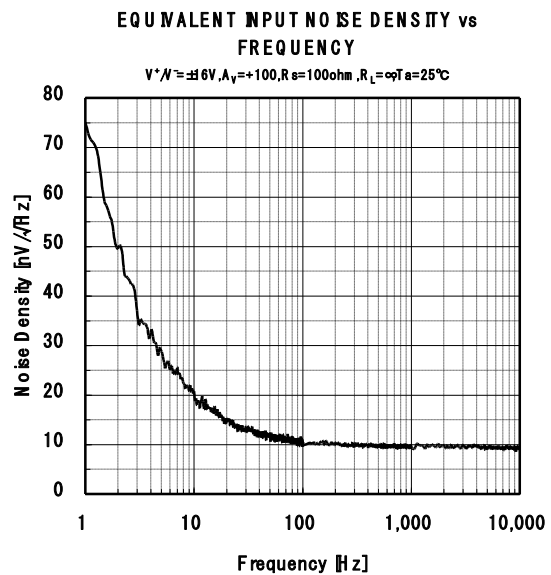
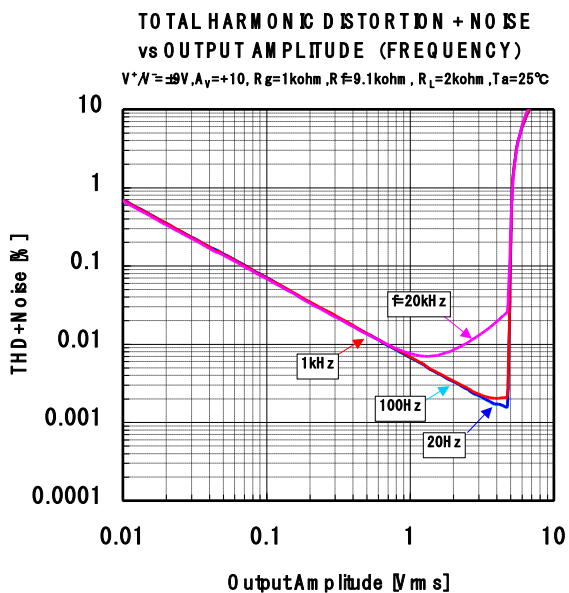
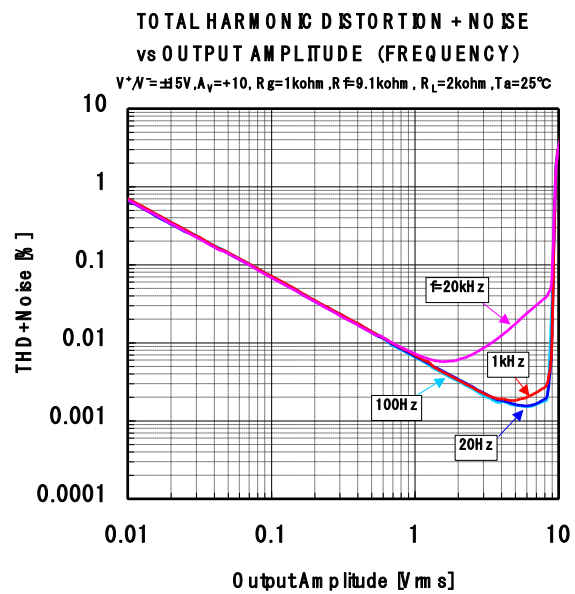
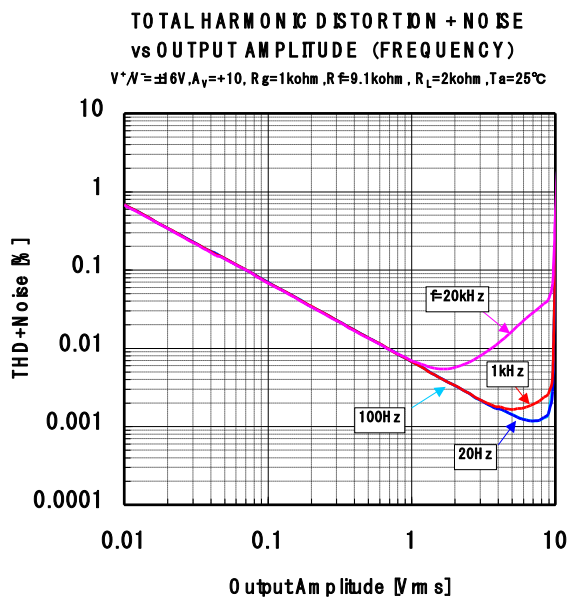
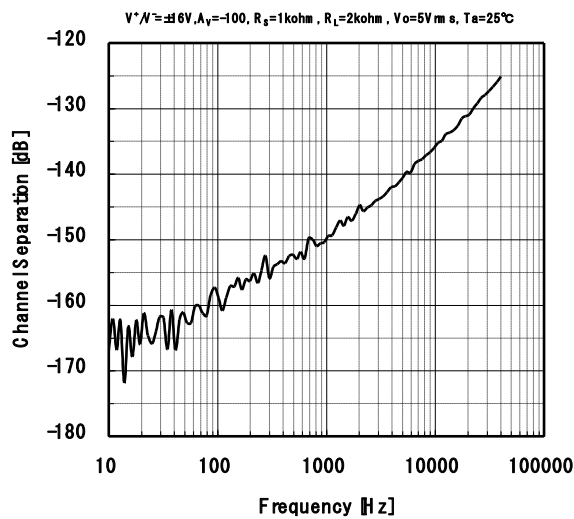


Fig.1 Power Dissipations vs. Ambient Temperature on the MUSES01

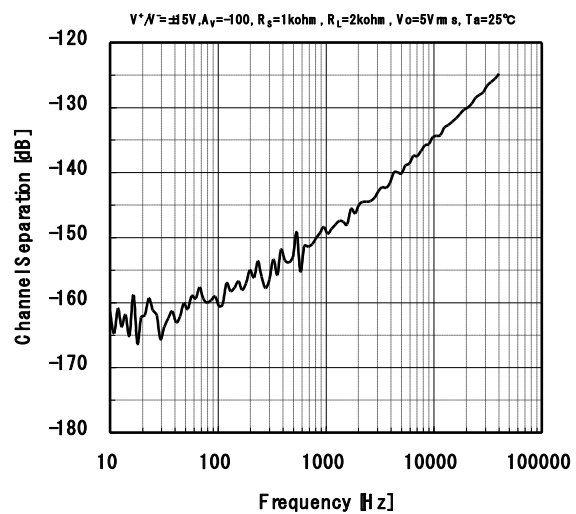
■ TYPICAL CHARACTERISTICS



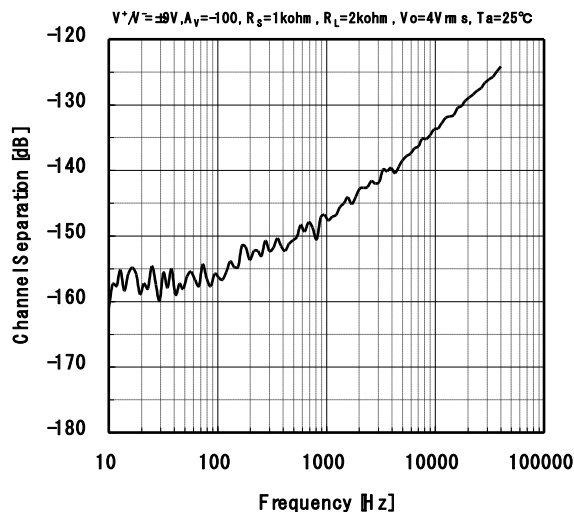
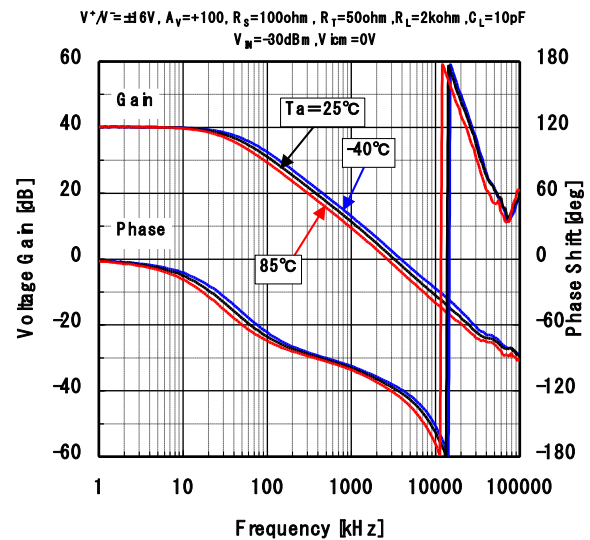
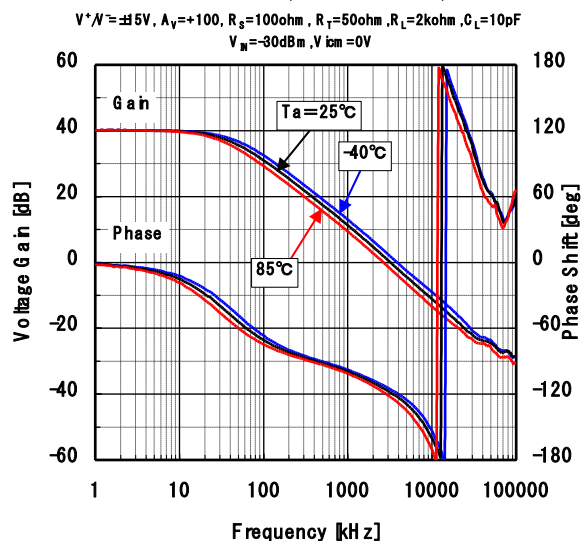
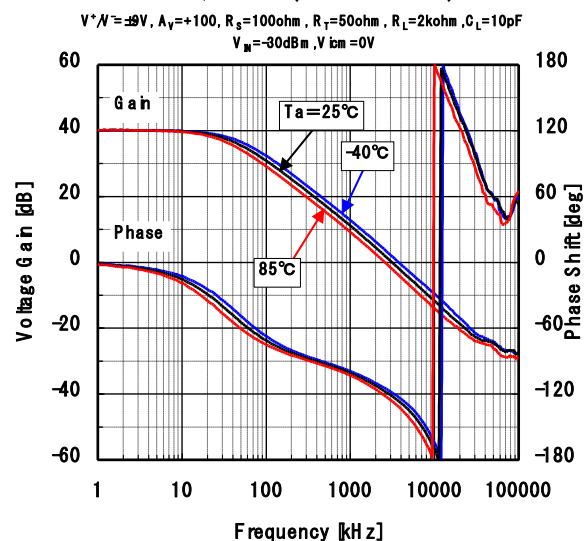
CHANNEL SEPARATION vs FREQUENCY



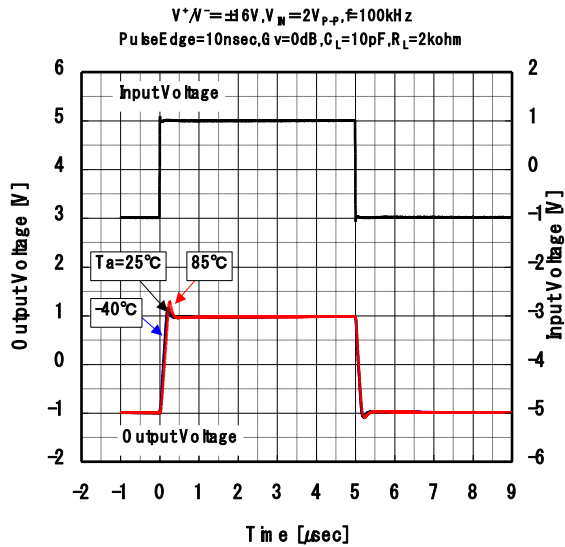
CHANNEL SEPARATION vs FREQUENCY



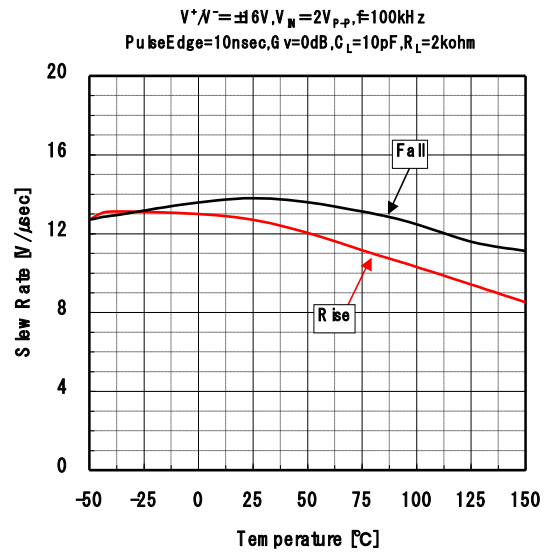
CHANNEL SEPARATION vs FREQUENCY

CLOSED-LOOP GAIN/PHASE vs
FREQUENCY (TEMPERATURE)CLOSED-LOOP GAIN/PHASE vs
FREQUENCY (TEMPERATURE)CLOSED LOOP GAIN/PHASE vs
FREQUENCY (TEMPERATURE)

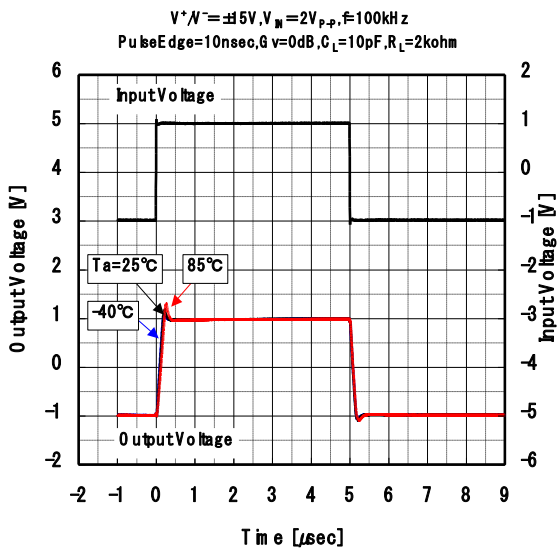
TRANSIENT RESPONSE (TEMPERATURE)



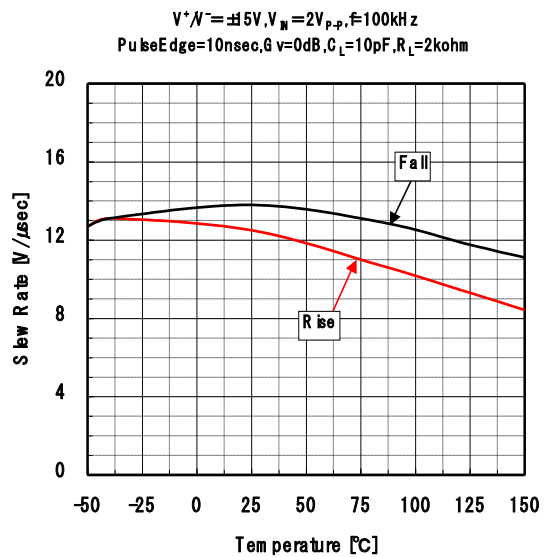
SLEW RATE vs TEMPERATURE



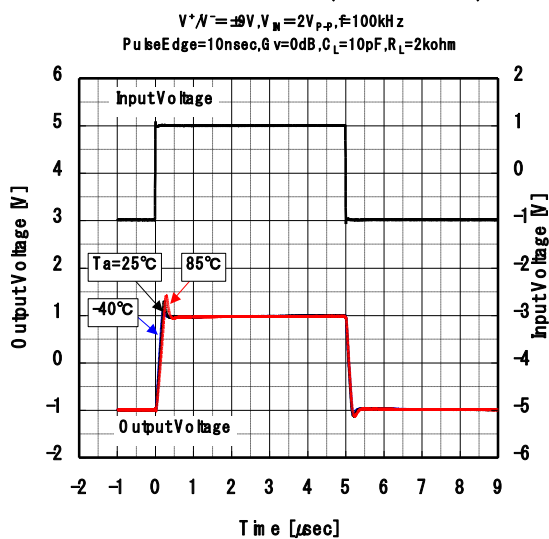
TRANSIENT RESPONSE (TEMPERATURE)



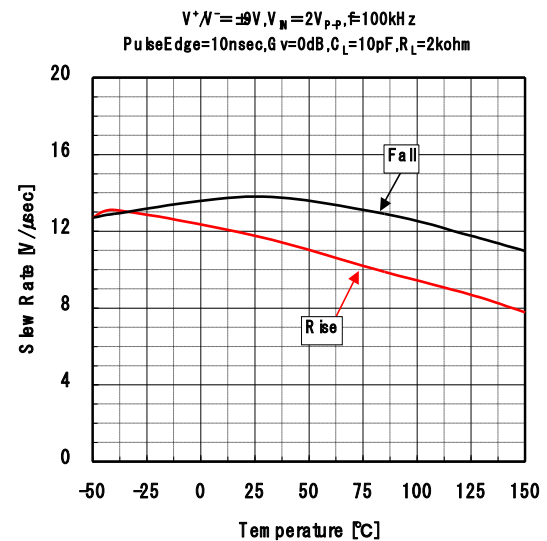
SLEW RATE vs TEMPERATURE

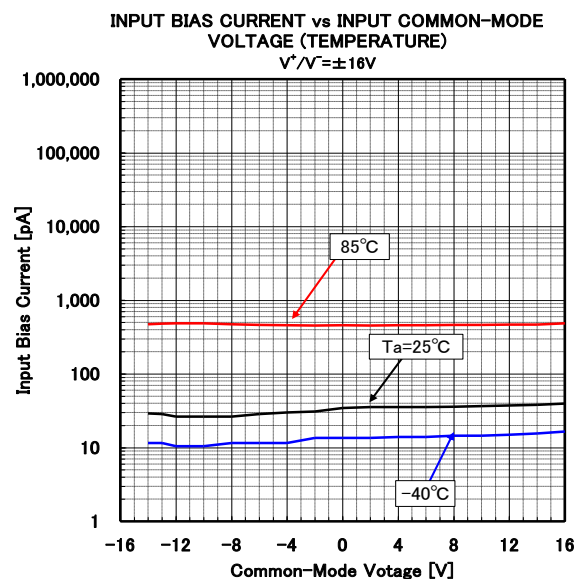
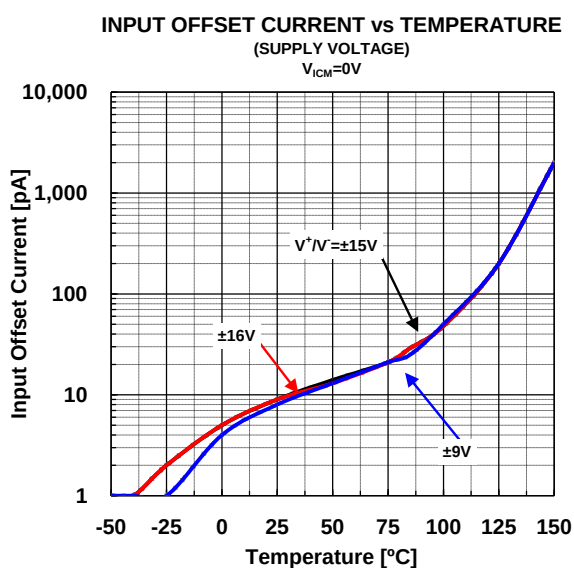
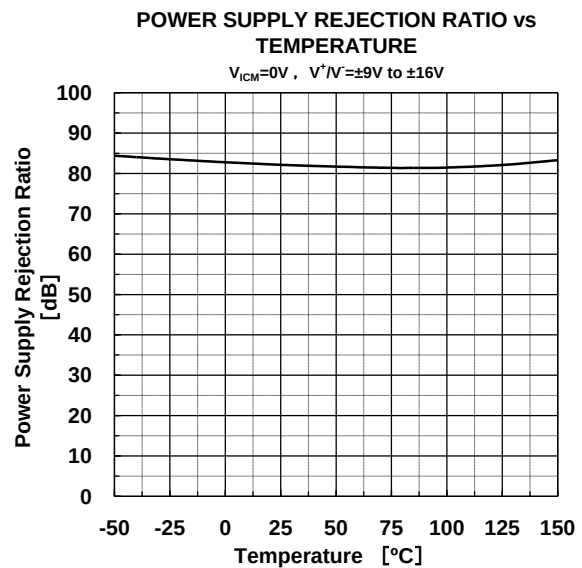
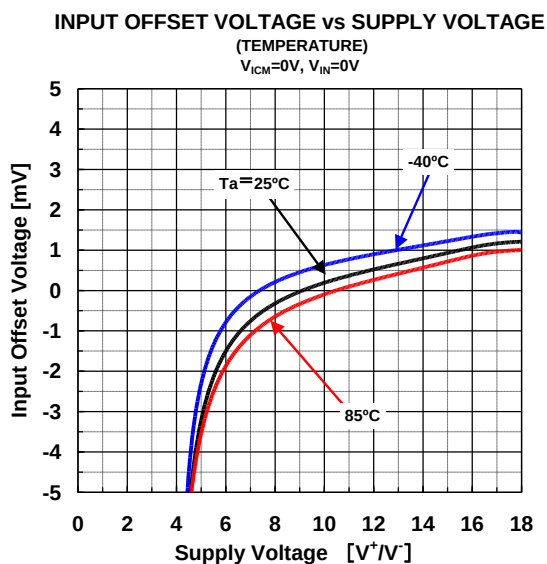
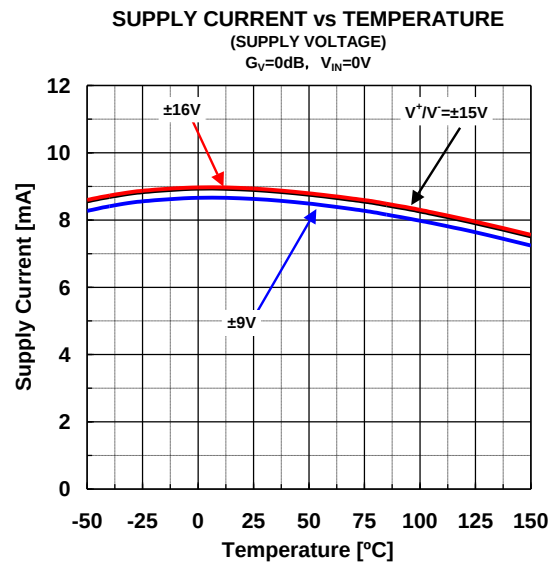
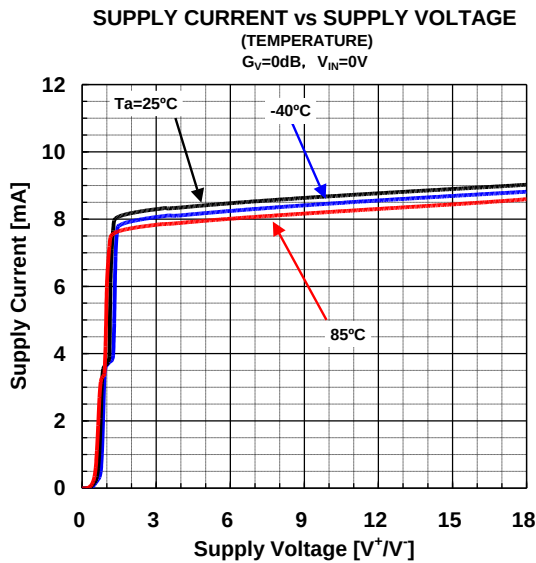


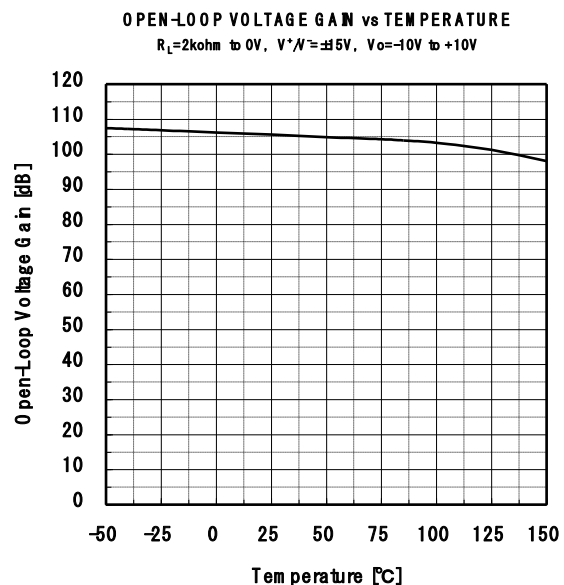
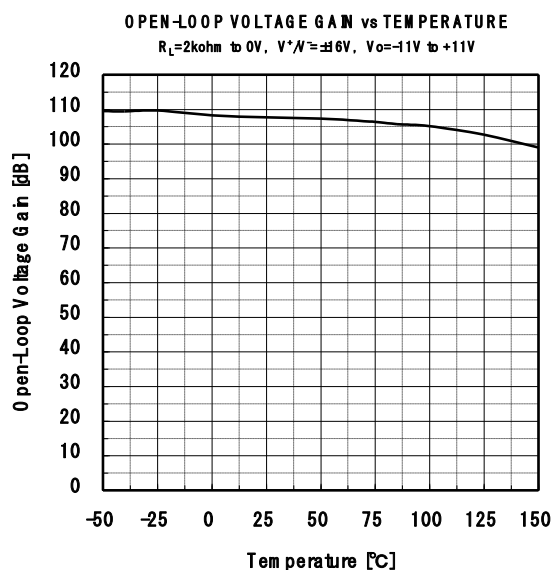
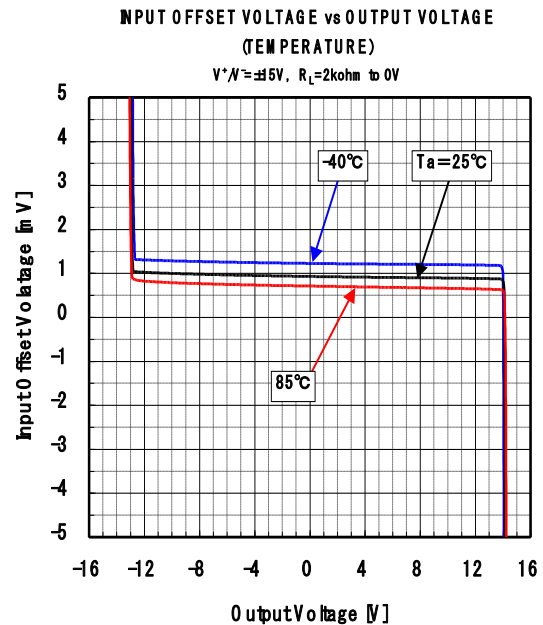
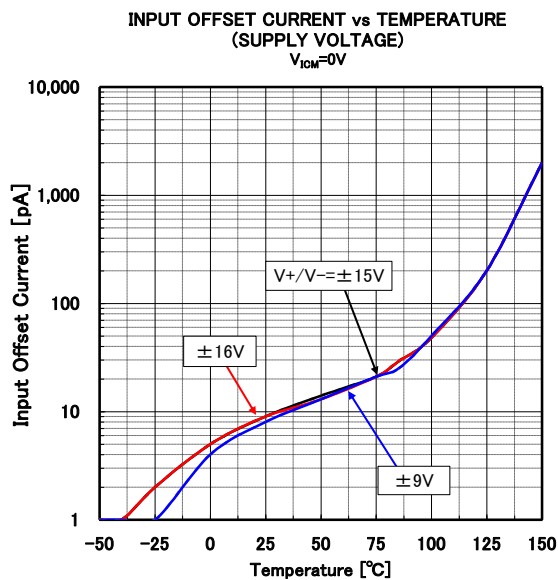
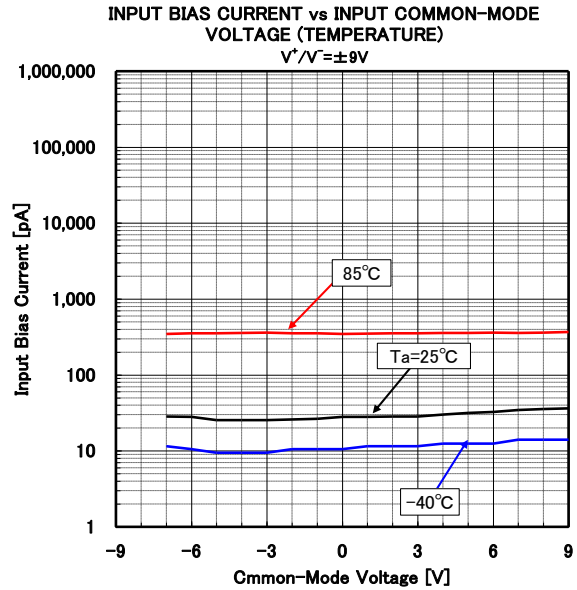
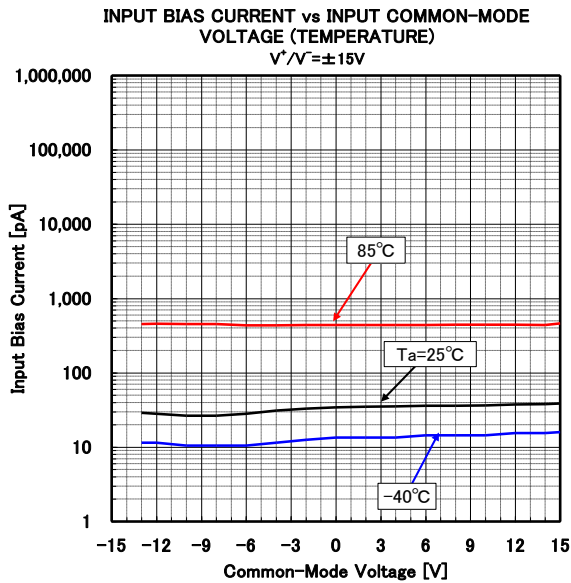
TRANSIENT RESPONSE (TEMPERATURE)

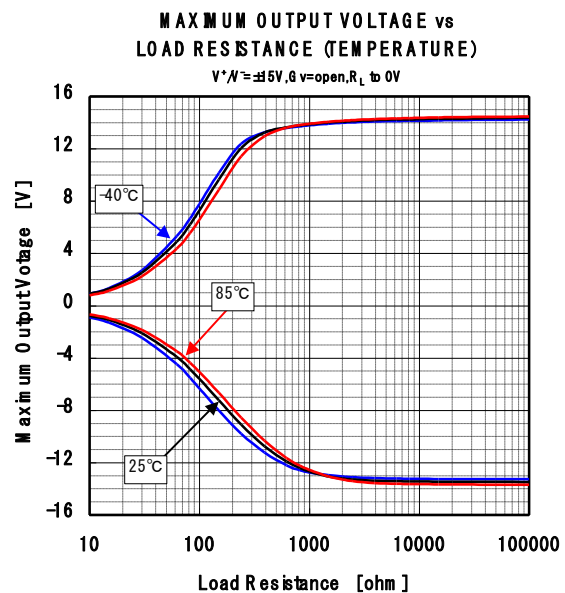
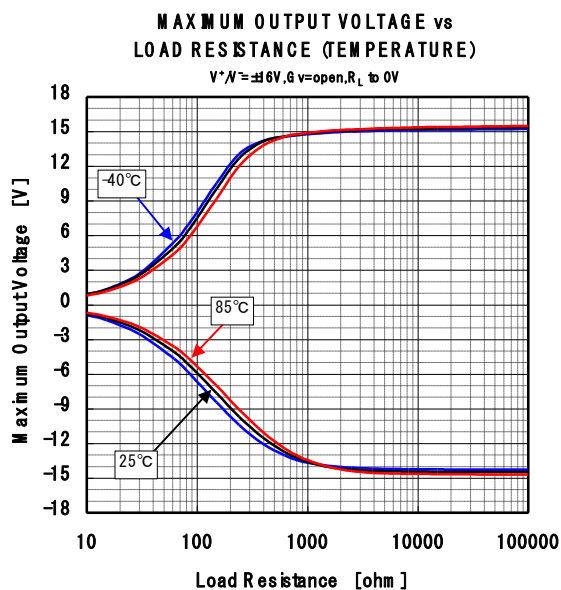
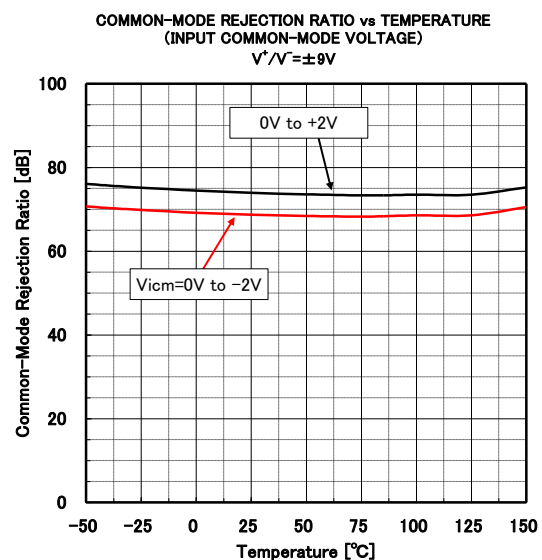
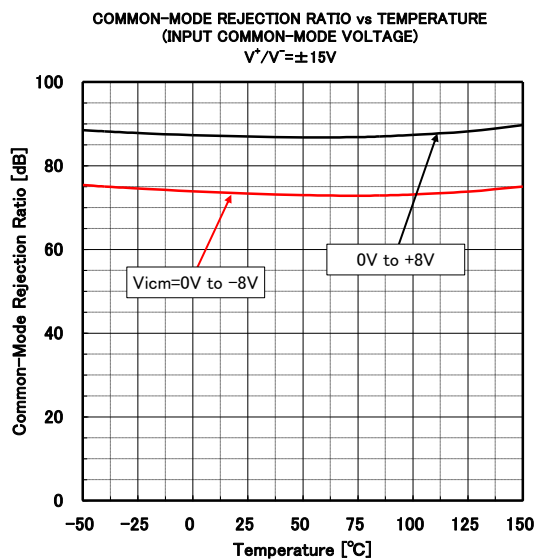
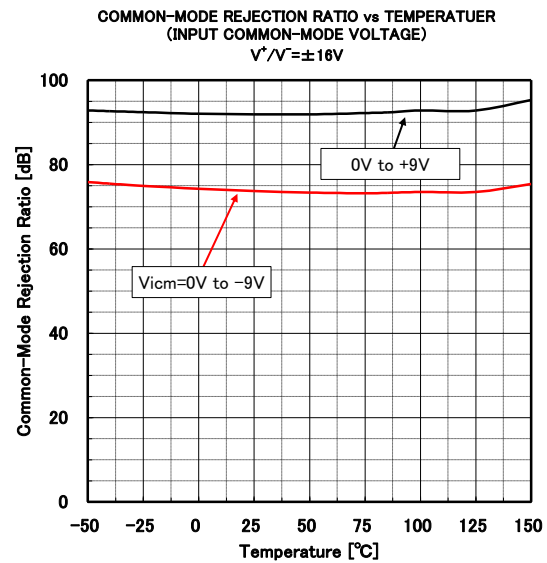
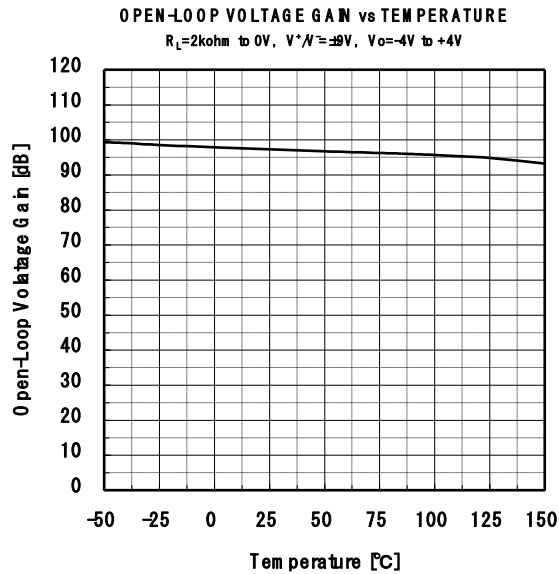


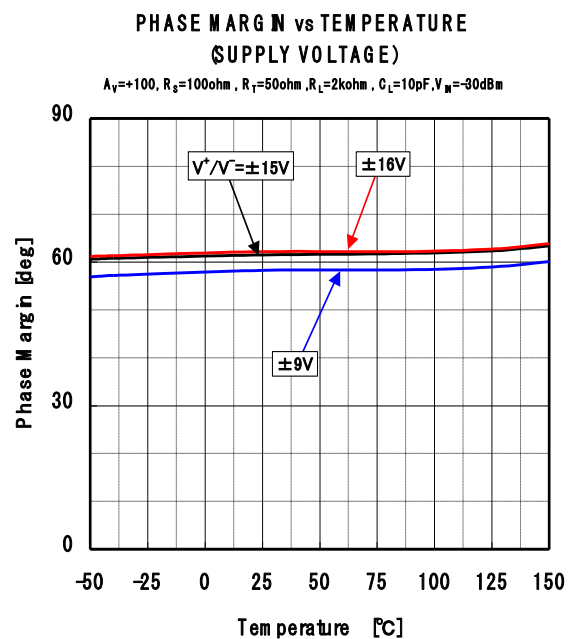
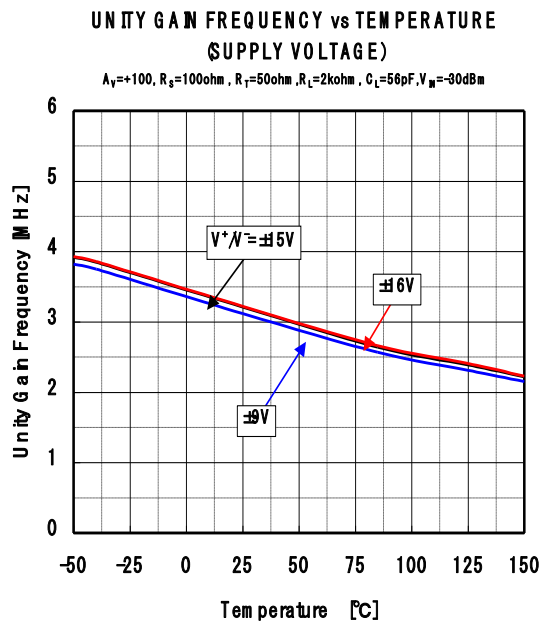
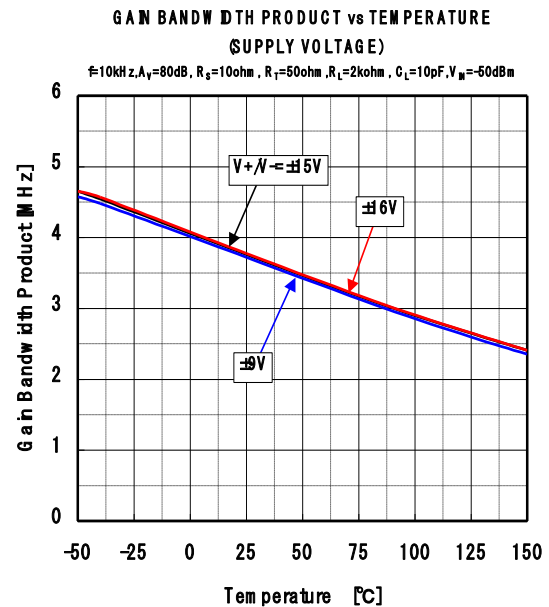
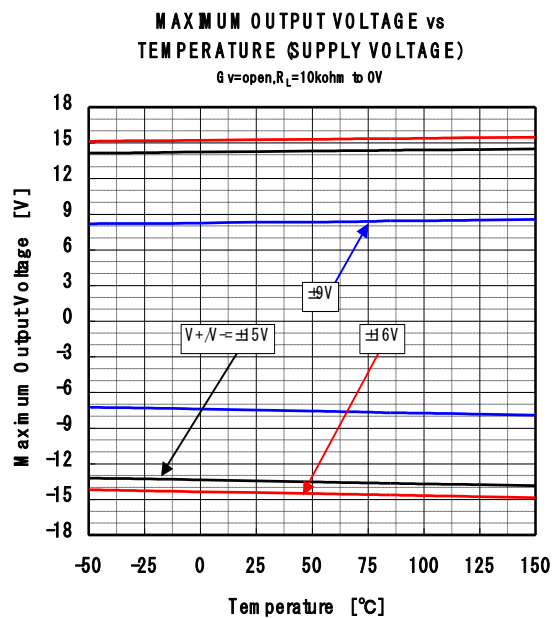
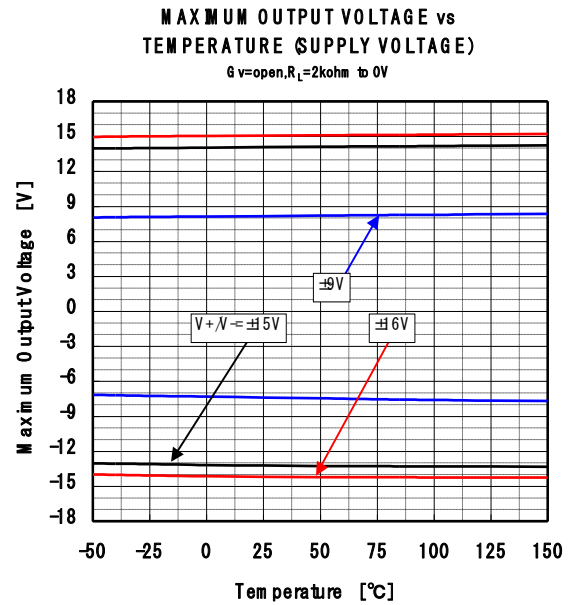
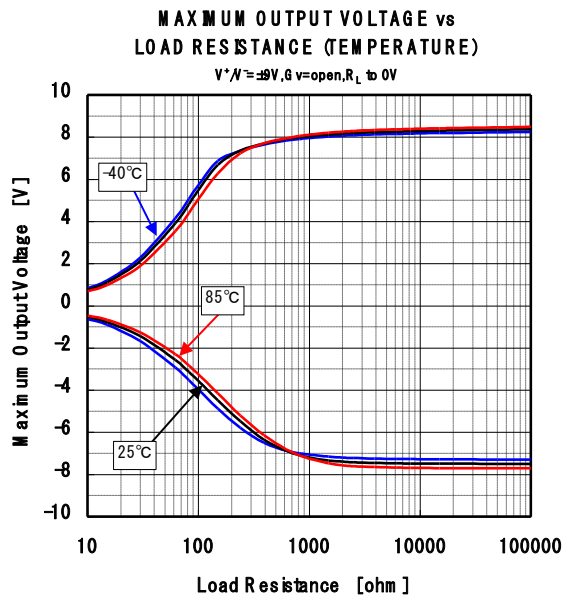
SLEW RATE vs TEMPERATURE











MEMO

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